Array of strings and its features

C – Array of Strings

In C programming String is a 1-D array of characters and is defined as an array of characters.

But an array of strings in C is a two-dimensional array of character types.

Each String is terminated with a null character (\0).

It is an application of a 2d array.

Syntax:

char variable_name[r][m] = {list of string};

where

- **var_name** is the name of the variable in C.
- **r** is the maximum number of string values that can be stored in a string array.
- **m** is the maximum number of character values that can be stored in each string array.

C – Array of Strings

```
// C Program to print Array of strings
#include <stdio.h>
int main()
{
   char arr[3][25] = {“Computer", “Computer Science", “Computer Applications"};
   printf("String array Elements are:\n");
   for (int i = 0; i < 3; i++)
   {
         printf("%s\n", arr[i]);
   }
   return 0;
 }
```

Output

```
String array Elements are:
Computer
Computer Science
Computer Applications
```

Memory is wasted as the array is prespecified and each array is given the max space even if it is not occupied.

We can’t directly change or assign the values to an array of strings in C.

Like **arr[0] = “abc"; // will create error**